map< const std::type_info *, std::vector < std::shared_ptr< smacc2::introspection::SmaccStateReactorInfo > > >
map< const std::type_info *, std::vector < std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
weak_ptr< smacc2::introspection::SmaccStateMachineInfo >
weak_ptr< smacc2::introspection::SmaccStateInfo >
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector < smacc2::introspection::ClientBehaviorInfoEntry > >
string
int
SharedPtr
SharedPtr
vector< smacc2::ISmaccState * >
shared_ptr< smacc2::introspection::SmaccStateInfo >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
shared_ptr< smacc2::introspection::SmaccStateMachineInfo >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
map< std::string, std::pair< std::function < std::string()>, boost::any > >
vector< smacc2_msgs::msg::SmaccTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
+stateReactorsInfo
+eventGeneratorsInfo
+stateMachine_
+parentState_
+transitions_
+children_
+tid_
+staticBehaviorInfo
+demangledStateName
+fullStateName
+depth_
+stateIndex_
-reactorWalkDepth_
#nh_
#timer_
#currentState_
#currentStateInfo_
#status_msg_
#orthogonals_
#stateMachineInfo_
-eventQueueMutex_
-m_mutex_
-stateMachineCurrentAction
-stateCallbackConnections
-globalData_
-transitionLogHistory_
-runMode_
-stateSeqCounter_
#eventsLogPub_
#stateMachinePub_
#stateMachineStatusPub_
#transitionHistoryService_
#transitionLogPub_
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::StateReactor > >
vector< std::shared_ptr< smacc2::SmaccEventGenerator > >
mutex
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::vector < std::shared_ptr< smacc2::ISmaccClientBehavior > > >
#node_
atomic< bool >
SmaccFifoScheduler *
processor_handle
thread
ExecutionModel
vector< smacc2::ISmaccUpdatable * >
vector< std::vector < smacc2::ISmaccUpdatable * > >
double
-statusPub_
#stateInfo_
#logger_
#stateReactors_
#eventGenerators_
-mutex_
#clients_
-clientBehaviors_
-stateMachine_
-smaccStateMachine_
-signalDetector_
-end_
-initialized_
-rosInitialized_
-scheduler_
-processorHandle_
-signalDetectorThread_
-executionModel_
-updatableClients_
-updatableStateElements_
-loop_rate_hz
smacc2::ISmaccState
+ ~ISmaccState()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionTypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
#parentState_
smacc2::ISmaccOrthogonal
+ ~ISmaccOrthogonal()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
boost::signals2::signal < Signature, optional_last_value< typename boost::function_traits< Signature >::result_type >, int, std::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost::signals2::mutex >
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
-stateMachine_
#components_
#stateMachine_
-currentState
-currentOrthogonal
-topicName_
-pub_
-orthogonal_
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()
boost::signals2::signal < void(ArrowKey arrow), optional_last_value< typename boost::function_traits< void(ArrowKey arrow) >::result_type >, int, std::less< int >, function< void(ArrowKey arrow)>, typename extended_signature < function_traits< void(ArrowKey arrow) >::arity, void(ArrowKey arrow)>::function_type, boost::signals2::mutex >
smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >
boost::signals2::signal < void(char keypress), optional_last_value< typename boost::function_traits< void (char keypress)>::result_type >, int, std::less< int >, function < void(char keypress)>, typename extended_signature< function_traits< void(char keypress)>::arity, void(char keypress)> ::function_type, boost::signals2::mutex >
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
#owner_
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()
< void(ArrowKey arrow)>
< void(char keypress)>
smacc2::SmaccSignal < void(ArrowKey arrow)>
smacc2::SmaccSignal < void(char keypress)>
function< void(std_msgs::msg::UInt16)>
bool
optional< rmw_qos_reliability_policy_t >
optional< int >
optional< rmw_qos_durability_policy_t >
-mutex_
smacc2::SmaccClientBehavior
+ ~SmaccClientBehavior()
+ onEntry()
+ onExit()
+OnArrowPress_
+OnKeyPress_
-postEventKeyPress
-initialized_
+reliability
+queueSize
+durability
cl_keyboard::components::CpKeyboardListener1
+ CpKeyboardListener1()
+ ~CpKeyboardListener1()
+ onComponentInitialization()
+ OnKeyPress()
+ OnArrowPress()
+ onKeyboardMessage()
+ postKeyEvent()
smacc2::client_core_components::CpTopicPublisher < MessageType >
+ CpTopicPublisher()
+ ~CpTopicPublisher()
+ publish()
+ onInitialize()
optional< rclcpp::Duration >
optional< rclcpp::Time >
-periodDuration_
-lastUpdate_
< geometry_msgs::msg::Twist >
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()
smacc2::client_core_components::CpTopicPublisher < geometry_msgs::msg::Twist >
+ CpTopicPublisher()
+ ~CpTopicPublisher()
+ publish()
+ onInitialize()
float
Twist
optional< std::chrono::steady_clock::time_point >
milliseconds
-keyboardListener_
-twistPublisher_
-angularStep_
-linearStep_
-heldTwist_
-idleTwist_
-lastArrowTime_
-deadman_
cl_keyboard::CbKeyboardTwistTeleop
+ CbKeyboardTwistTeleop()
+ onEntry()
+ onExit()
+ update()
- onArrowPress()